

■ INTRODUCTION

The CE6211 Series are a group of dual channel low-dropout voltage regulators designed for portable and wireless applications that require high PSRR, low quiescent current and excellent line and load transient response. The CE6211 includes a reference voltage source, error amplifiers, driver transistors, current limiters and phase compensators internally. The CE6211 is stable with a small 1 μ F ceramic on the output, which is ideal for battery powered systems for delivering low dropout voltage and low quiescent current. It provides up to 300mA at each channel, from a 2.0V to 6.0V input. The CE6211 is available in 6 pin SOT-23 package.

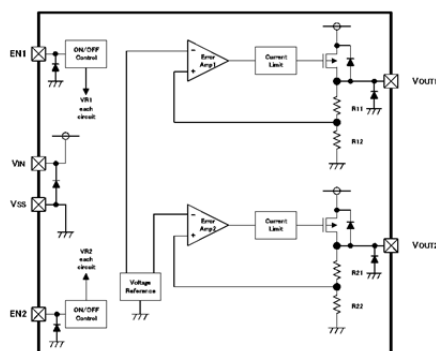
■ FEATURES

- Shutdown Current: < 0.1 μ A
- Output Current: 300mA
- Output Voltage Range: 1.2V ~ 5.0V,
(selectable in 0.1V steps)
- High Accuracy: $\pm 2\%$ (Typ.)
- Dropout Voltage:
100mV@100mA (3.0V Typ.)
- Excellent Line Regulation: 0.01%/V
- Built-in Current Limiter
- Built-in Short Circuit Protection
- Static safety: 2KV@HBM
- TC: 100ppm/ $^{\circ}$ C
- Low ESR Ceramic Capacitor Compatible

■ APPLICATIONS

- Mobile phones
- WLAN and Bluetooth appliances
- Portable Audio Equipments
- Cordless telephone
- Cameras, Video recorders
- Battery powered portable devices

■ BLOCK DIAGRAM



■ ORDER INFORMATION

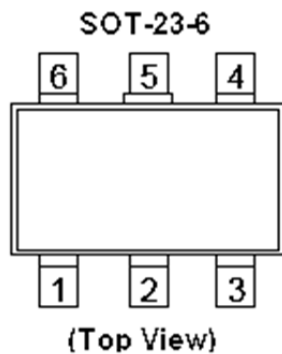
CE6211①②③④⑤

DESIGNATOR	SYMBOL	DESCRIPTION
①②	EE	High Active
③④	I II	I :Regulator1 Output Voltage II : Regulator2 Output Voltage (See Chart1)
⑤	E	Package: SOT-23-6

Chart1 (Regulator1, Regulator2 Output Voltage Indicate)

symbol	Output Voltage	symbol	Output Voltage	symbol	Output Voltage	symbol	Output Voltage
a	0.9	A	3.5	n	2.2	N	4.8
b	1.0	B	3.6	o	2.3	O	4.9
c	1.1	C	3.7	p	2.4	P	5.0
d	1.2	D	3.8	q	2.5	Q	5.1
e	1.3	E	3.9	r	2.6	R	5.2
f	1.4	F	4.0	s	2.7	S	5.3
g	1.5	G	4.1	t	2.8	T	5.4
h	1.6	H	4.2	u	2.9	U	5.5
i	1.7	I	4.3	v	3.0	V	5.6
j	1.8	J	4.4	w	3.1	W	5.7
k	1.9	K	4.5	x	3.2	X	5.8
l	2.0	L	4.6	y	3.3	Y	5.9
m	2.1	M	4.7	z	3.4	Z	6.0

■ PIN CONFIGURATION



PIN NUMBER	SYMBOL	FUNCTION
1	EN1	ON/OFF Control 1
2	V _{IN}	Power Input
3	En2	ON/OFF Control 2
4	V _{OUT2}	Output 2
5	V _{SS}	Ground
6	V _{OUT1}	Output 1

■ ABSOLUTE MAXIMUM RATINGS

(Unless otherwise specified, Ta=25°C)

PARAMETER		SYMBOL	RATINGS	UNITS
Input Voltage		V _{IN}	V _{SS} -0.3~V _{SS} +7	V
Output Current		I _{OUT1} + I _{OUT2}	700	mA
Output Voltage		V _{OUT}	V _{SS} -0.3~V _{IN} +0.3	V
Power Dissipation	SOT-23-6	Pd	400	mW
Operating Temperature		T _{opr}	-40~+85	°C
Storage Temperature		T _{stg}	-40~+125	°C
Soldering Temperature & Time		T _{solder}	260°C, 10s	

■ ELECTRICAL CHARACTERISTICS

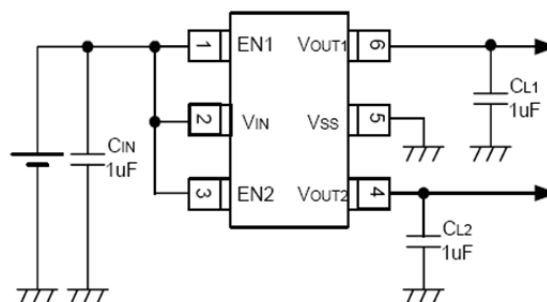
Regulator1, Regulator2 ($V_{IN}=V_{OUT}+1V$, $C_{IN}=C_{OUT}=1\mu F$, $T_a=25^\circ C$, unless otherwise specified)

PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
Output Voltage	$V_{OUT(E)}$ (Note 2)	$I_{OUT}=40mA$, $V_{IN}=V_{OUT}+1V$, $V_{IN}\geq 2V$	$1.5V < V_{OUT} \leq 5.0V$	$V_{OUT} * 0.98$	V_{OUT} (Note 1)	$V_{OUT} * 1.02$	V
			$1.2V \leq V_{OUT} \leq 1.5V$	$V_{OUT} - 0.03$	V_{OUT}	$V_{OUT} + 0.03$	V
Supply Current	I_{SS}	$V_{EN}=V_{IN}=V_{OUT}+1V$			100	160	μA
Shutdown Current	I_{SHDN}	$V_{EN} = V_{SS}$			0.1	1.0	μA
Output Current	I_{OUT}	$V_{IN}\geq 2V$, $V_{IN}=V_{OUT}+1V$	300				mA
Dropout Voltage (Note 3)	V_{dif}	$I_{OUT} = 100mA$ $V_{OUT}\geq 2.8V$			100		mV
Load Regulation	ΔV_{OUT}	$V_{IN}= V_{OUT} + 1V$, $1mA \leq I_{OUT} \leq 100mA$			10		mV
Line Regulation	$\frac{\Delta V_{OUT}}{\Delta V_{IN} * V_{OUT}}$	$I_{OUT} = 10mA$ $V_{OUT} + 1V \leq V_{IN} \leq 6V$			0.01	0.2	%/V
Output Voltage Temperature Characteristics	$\frac{\Delta V_{OUT}}{\Delta T * V_{OUT}}$	$I_{OUT} = 10mA$ $-40 \leq T \leq +85$			100		ppm/ $^\circ C$
Power Supply Ripple Rejection	PSRR	$I_{OUT} = 50mA$ $f = 1KHz$			70		dB
Short Current	I_{Short}	$V_{OUT} = V_{SS}$			100		mA
Current Limit	I_{Lim}	$V_{IN}= V_{OUT} + 1V$			600		mA
Input Voltage	V_{IN}	—	2.0			6.0	V
CE "High" Voltage	V_{CE} "H"		1.5			V_{IN}	V
CE "Low" Voltage	V_{CE} "L"					0.3	V

NOTE:

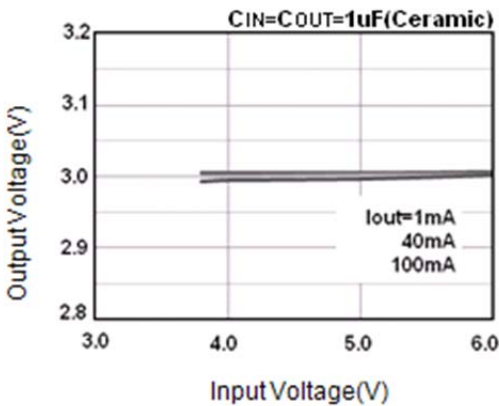
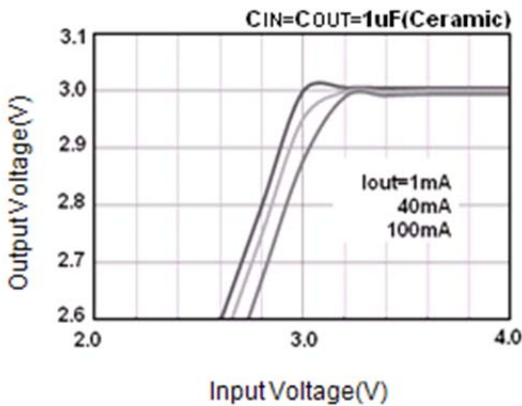
1. V_{OUT} : Specified Output Voltage.
2. $V_{OUT} (E)$: Effective Output Voltage (I.e. The Output Voltage When $V_{IN} = (V_{OUT} + 1.0V)$ And Maintain A Certain I_{OUT} Value).
3. V_{dif} : The Difference Of Output Voltage And Input Voltage When Input Voltage Is Decreased Gradually Till Output Voltage Equals To 98% Of $V_{OUT} (E)$; When $V_{OUT} < 2.0V$, $V_{IN} \geq 2.0V$ Should be Guaranteed.

■ TYPICAL APPLICATION

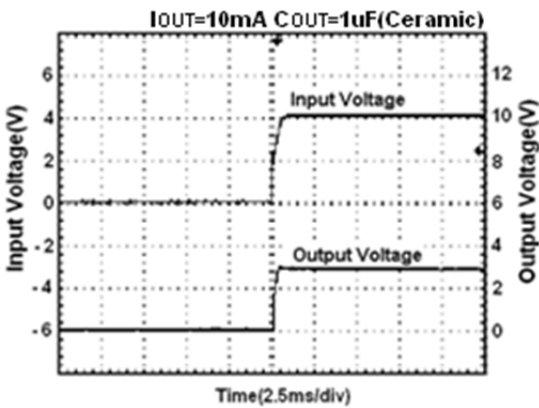
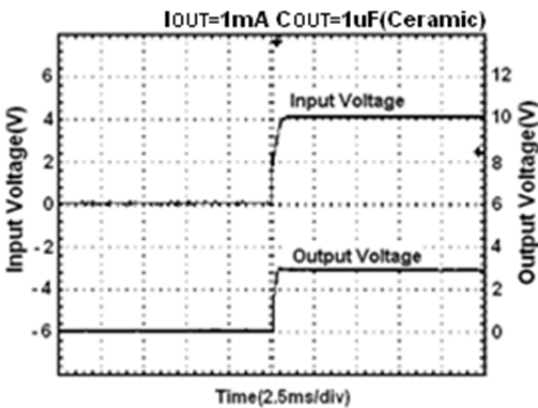


■ TYPICAL PERFORMANCE CHARACTERISTICS

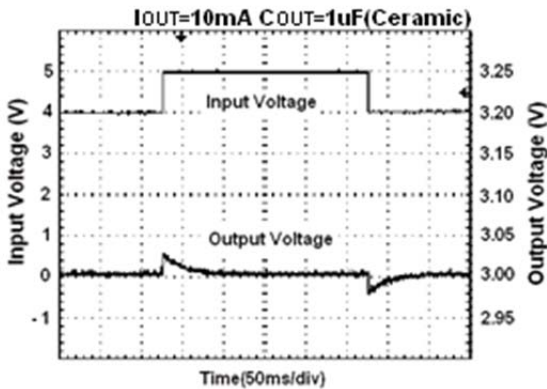
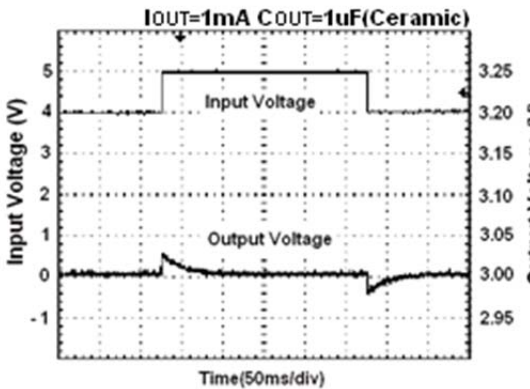
(1) Output Voltage vs. Input Voltage



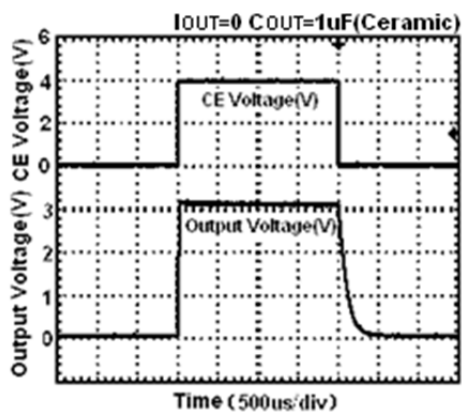
(2) Input Transient Response 1



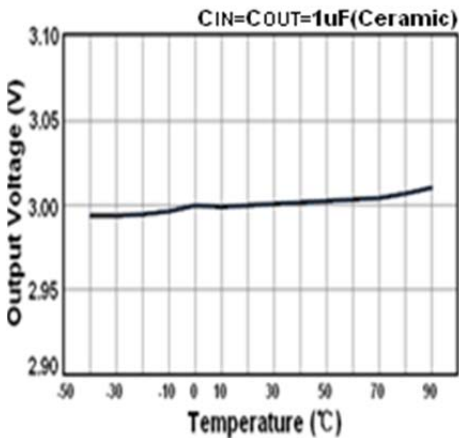
(3) Input Transient Response 2



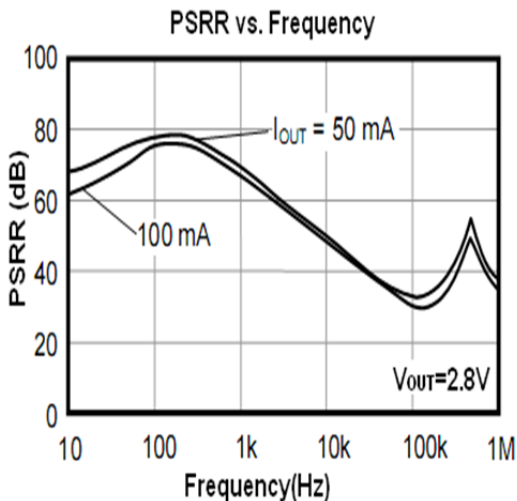
(4) EN Shutdown Response



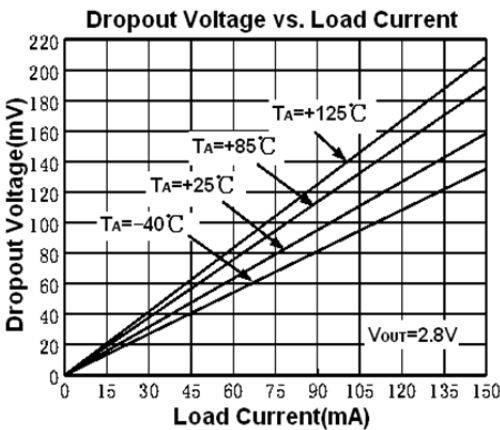
(5) Output Voltage vs. Temperature



(6) PSRR

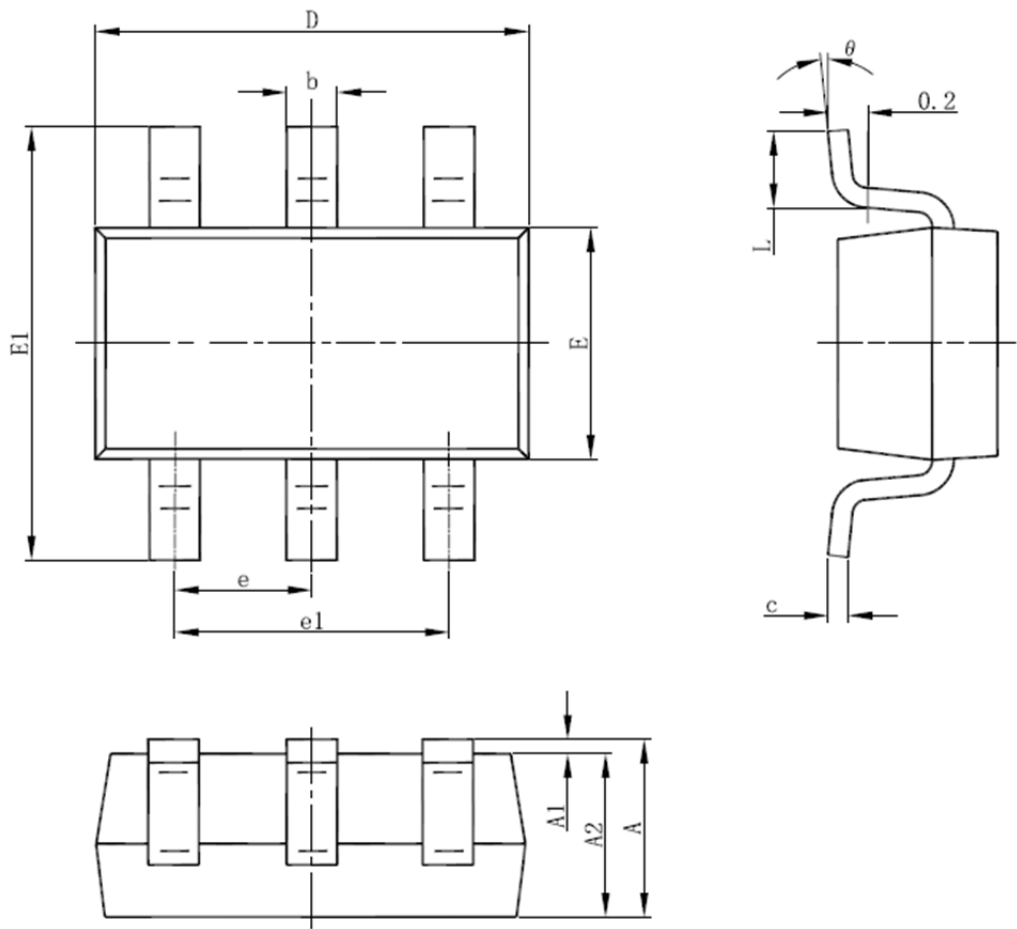


(7) Dropout



■ PACKAGING INFORMATION

● SOT-23-6 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E	1.500	1.700	0.059	0.067
E1	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
θ	0°	8°	0°	8°

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